

Interrelations of Current Health Expenditure and Fertilizer Consumption in European Countries: Step-by-Step to Sustainable Agricultural Practices

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Abstract: *This research explores interrelations between fertilizer consumption and health expenditure in European countries examining how agricultural practices impact public health costs through pollution-related risks. Use of fertilizers, particularly nitrogen-based ones, is linked to environmental pollution (such as nitrate contamination in water and greenhouse gas emissions), which has well-documented health impacts. This study aims to investigate whether higher levels of fertilizer consumption correlate with increased healthcare spending, reflecting the economic burden of pollution-related health conditions. Via panel data from 37 European countries between 2006 and 2021, the panel regression analysis with the Fixed Effect, Random Effect, and Pooling Models is applied to evaluate relationships between fertilizer consumption and health expenditure per capita and as a percentage of GDP. The Fixed Effect Model, supported by the Hausman test, was ultimately selected for its ability to control for country-specific factors. The results indicate that higher fertilizer use is associated with increased health expenditure per capita, underscoring the economic impact of pollution-related illnesses. Additionally, food import shares show a positive association with health expenditure. It suggests that reliance on imported food might contribute to higher healthcare costs due to potential health risks from imported food quality. The export share effect on health expenditure is more complex, with mixed results across models, indicating both potential economic benefits and environmental health costs. These findings emphasize need for sustainable agricultural policies that balance productivity with public health. By revealing the financial implications of intensive fertilizer use, this research underscores the importance of integrating preventative measures (such as minimizing fertilizer run-off and promoting eco-friendly farming practices) into policy frameworks. This study provides critical data to support the European Green Deal and Sustainable Development Goals.*

Keywords: fertilizer consumption, health expenditure, panel regression analysis, public health, sustainable agriculture

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INTRODUCTION

Interrelations of current health expenditure and fertilizer consumption are relevant and timely for several reasons. They concern public health, environmental problems, and agricultural practices.

Fertilizer consumption is closely linked to environmental concerns like soil degradation, water contamination, and greenhouse gas emissions (GGE). In Europe, where there are strict environmental regulations, understanding the impact of fertilizer use on public health can help policymakers balance agricultural productivity with environmental sustainability.

According to the European Environment Agency report on nitrate pollution in European waters (EEA, 2018), fertilizer use is a major source of nitrate contamination in water bodies. Excessive application of nitrogen-based fertilizers leads to nutrient run-off into rivers, lakes, and coastal waters. It contributes to water pollution and eutrophication (over-enrichment with nutrients), which negatively affects aquatic ecosystems and can cause toxic algal blooms.

The contamination of water with nitrates is not only harmful to the environment but also poses significant public health risks. High nitrate levels in drinking water have been associated with health issues, including an increased risk of methemoglobinemia (blue baby syndrome) in newborns and potential relations to evident cancers. The EU Nitrates Directive (European Commission, 2021) aims to reduce water pollution by regulating nitrogen use and monitoring nitrate concentrations in water sources. Understanding the public health risks linked to fertilizer-driven nitrate pollution helps policymakers enforce and adapt these regulations to balance agricultural needs with public health priorities.

In this context, examining how fertilizer use correlates with health expenditure can reveal if regions with high fertilizer consumption also bear higher healthcare costs related to environmental health risks. Fertilizer use, particularly nitrogen fertilizers, contributes to GGE, specifically nitrous oxide (N_2O), a potent GGE with a global warming potential approximately 300 times greater than carbon dioxide (CO_2). According to the Intergovernmental Panel on Climate Change (IPCC, 2019) and various studies on European agriculture, nitrous oxide emissions from agricultural soils are primarily driven by nitrogen-based fertilizer application and other soil management practices. These emissions contribute to climate change, which has both environmental and health implications.

The GGEs from fertilizers exacerbate climate change, which, in turn, affects public health through extreme weather events, food security risks, and changing disease patterns. Recognizing these impacts, the European Farm to Fork Strategy (European Commission, 2020) aims to reduce fertilizer use by 20% and promote sustainable agricultural practices. By understanding the environmental and health impacts of fertilizer use, European policymakers are better equipped to set realistic targets for reducing emissions while maintaining agricultural productivity, thereby aligning environmental sustainability goals with public health outcomes.

Fertilizers play a crucial role in enhancing agricultural productivity, which is vital for food security. In Europe, where land is limited, efficient use of fertilizers can boost yields and support food security. However, there is an ongoing debate about the trade-offs between maximizing agricultural productivity and maintaining environmental health.

Food and Agriculture Organization report (FAO, 2019) emphasizes that fertilizers are essential for improving crop yields and achieving food security, particularly in regions with limited arable lands, such as Europe. It discusses how increased fertilizer use has allowed countries to meet food demands despite land limitations, especially in developed regions. However, it also highlights the trade-offs, noting that unsustainable fertilizer use can degrade soil health and pollute water sources, thereby posing risks to long-term food security and environmental health.

The Farm to Fork Strategy within the European Green Deal (European Commission, 2020) acknowledges that fertilizers are essential to maintaining agricultural productivity in Europe, where arable land is scarce. However, the strategy also addresses the environmental concerns associated with fertilizer overuse, such as soil degradation and water pollution. It advocates for efficient fertilizer management to support food security and minimize environmental impacts underscoring the balance needed between productivity and environmental sustainability.

Health expenditure and agricultural practices are often interrelated in complex ways. Higher agricultural productivity might improve national income, potentially leading to greater investment in healthcare. Exploring these relationships could provide insights into how agricultural and health policies are connected in European economies. The World Bank (2020) discusses how increased agricultural productivity contributes to economic growth and, consequently, higher national income. This increase in national income

allows governments to invest more in essential sectors, including healthcare. Higher agricultural productivity in European economies can indirectly lead to enhanced health expenditure by improving national budgets, which can then be allocated toward better health infrastructure and services. This relationship suggests that agricultural productivity not only supports food security but can also promote better public health outcomes.

Regions with intensive fertilizer use may experience environmental pollution that leads to increased health risks and, consequently, higher healthcare costs. Studying this relationship in European countries, where health data and environmental policies are well-documented, can provide evidence of how agricultural practices influence public health spending.

The EEA report (2019) highlights significant health risks posed by environmental pollution in Europe, including pollution from agricultural sources like fertilizers. Excessive fertilizer use leads to nitrate contamination in water bodies and soil, contributing to health issues such as methemoglobinemia, certain cancers, and respiratory problems due to ammonia emissions. The report emphasizes that these pollution-related health risks translate into higher healthcare costs because public health systems in affected regions face the burden of treating pollution-related illnesses. It recommends sustainable agricultural practices as a key intervention to reduce pollution and alleviate associated healthcare costs.

Understanding links between fertilizer use and health expenditure can inform policies focused on preventative healthcare. If higher fertilizer consumption is associated with increased health spending due to pollution-related illnesses, countries could prioritize preventative measures (reducing fertilizer runoff, promoting organic farming) to lower long-term healthcare costs.

The WHO report (2015) emphasizes the importance of preventative healthcare policies aimed at reducing environmental risks, including pollution from agricultural sources. It highlights how preventative actions (such as minimizing fertilizer run-off and promoting eco-friendly farming practices) can mitigate pollution-related health issues and decrease healthcare costs. The report advocates for integrating environmental considerations into public health policies, as reducing exposure to environmental pollutants can significantly lower the burden on healthcare systems over time. In Europe, where data on pollution and health is well-documented, such policies could be particularly impactful.

The Farm to Fork Strategy under the European Green Deal (European Commission, 2020) aims to promote sustainable agricultural practices, including fertilizer use reduction and organic farming encouragement. By decreasing fertilizer run-off and other sources of pollution, the strategy seeks to lower the risk of pollution-related illnesses and associated healthcare costs. The European Commission highlights agricultural preventative measures can play a role in improving public health outcomes and reducing long-term healthcare expenses. This aligns with a need for policies that prioritize health by addressing root causes of pollution-related health risks in the agricultural sector.

The European Union has specific policies that regulate fertilizer use, including the EU Nitrates Directive and the Common Agricultural Policy (CAP). These regulations aim to reduce environmental pollution and support agricultural productivity. Investigating the impact of these policies on health expenditure can provide feedback on effectiveness of EU regulations in achieving sustainable outcomes. The European Green Deal sets ambitious targets to reduce pollution, including from agriculture. By exploring how fertilizer consumption impacts health expenditure, this research can contribute to the Green Deal goal of creating a healthier, more sustainable Europe. It may also highlight areas where further regulatory adjustments are necessary.

With advancements in technology, European countries are increasingly adopting precision agriculture techniques that optimize fertilizer use, potentially reducing its negative environmental impact. Studying how these practices affect health expenditure could show benefits of modern farming methods in both agricultural efficiency and public health.

The European Parliament study (2016) discusses the precision agriculture adoption across Europe and how it can optimize the use of inputs like fertilizers and pesticides. Precision agriculture techniques, such as soil monitoring, GPS-guided machinery, and data-driven application of fertilizers, help farmers apply the exact quantity of needed fertilizers, reducing excess and preventing run-off. By lowering the environmental impact, precision agriculture could potentially lead to lower pollution-related health risks, indirectly reducing healthcare costs associated with pollution. The report highlights that optimizing input use not only increases agricultural efficiency but also aligns with Europe's environmental and public health goals.

There is a growing trend in Europe toward organic and sustainable farming, which uses minimal or no synthetic fertilizers. This shift could impact public health and healthcare costs differently from conventional agriculture. Investigating these trends would help evaluate benefits of organic practices for both health and

environment. The European Commission's Organic Action Plan outlines the EU commitment to expanding organic farming across member states, with a target of 25% of agricultural land under organic farming by 2030. Organic farming practices avoid synthetic fertilizers and pesticides, which reduces soil and water pollution and promotes biodiversity. Studies referenced in the plan suggest that organic farming has potential health benefits due to lower chemical exposure and improved soil health, which indirectly supports public health by reducing pollution-related diseases. The plan emphasizes the dual environmental and health benefits of organic farming, suggesting that an increase in organic practices could reduce pollution-related healthcare costs over time (European Commission, 2020).

This topic is significant to Sustainable Development Goal 3, which targets to make certain healthy lives and uphold well-being for all. Understanding agricultural practices and health impact of fertilizer use can help European countries make progress toward this goal by addressing root causes of health issues. Sustainable agriculture, which includes responsible fertilizer use, is essential for achieving Sustainable Development Goal 2. Investigating interrelations between fertilizer use and health expenditure can contribute to strategies for sustainable food production without compromising public health.

Exploring this topic could help policymakers, researchers, and practitioners:

- Understand trade-offs between agricultural productivity and public health;
- Inform sustainable agricultural practices that minimize environmental health risks;
- Align agricultural and health policies with broader European goals, including the European Green Deal and Sustainable Development Goals.

This research has a potential to make a significant contribution to evidence-based policies that promote both economic development and public health, making it a valuable and timely area of study.

LITERATURE REVIEW

Agricultural practices, especially fertilizer use, have long been associated with both economic growth and environmental health challenges. In Europe, where land is limited and agricultural productivity is essential, fertilizers are critical for supporting crop yields and food security. However, fertilizers, particularly nitrogen-based types, contribute significantly to environmental pollution, which can have direct and indirect impacts on public health. As environmental and public health concerns intersect, there is a growing interest in understanding how these factors influence health expenditures in European economies.

The review paper by Snyder et al. (2009) discusses the impact of fertilizer use on GGEs, particularly nitrous oxide emissions from agricultural soils, and offers recommendations for managing fertilizers to reduce these emissions. The paper highlights the contribution of fertilizer use to climate change, which indirectly affects public health.

Study by Ward et al. (2018). analyzes health impacts of nitrate contamination in drinking water, including methemoglobinemia and potential links to cancer. They highlight public health risks of nitrate pollution stemming from agricultural fertilizers.

The article by Schneider and Gugerty (2011) examines the pathways through which agricultural productivity impacts broader economic development, including improved health outcomes. By increasing household and national income, agricultural productivity can reduce poverty and support public investments in healthcare and education. In European economies, efficient agricultural practices contribute to economic stability. It can allow for increased public spending on healthcare with a positive feedback loop between agricultural productivity, economic growth, and health expenditure.

The research by Schimmelpfennig (2016), though focused on the US, notes precision agriculture boosts profitability by lowering input costs like fertilizer while enhancing yields. These benefits could also be applied in Europe where stricter environmental regulations exist. Efficient fertilizer use in precision agriculture reduces pollution and related health risks, supporting Europe's environmental and public health objectives.

The paper by Mie et al. (2017) reviews health impacts of organic farming and food. Organic practices, which avoid synthetic fertilizers, can reduce exposure to harmful chemicals and associated health risks, like endocrine disruption and some cancers. They suggest that expanding organic farming in Europe could lower healthcare costs by reducing public exposure to pollutants from conventional agriculture, meeting public demand for cleaner, sustainable food, and benefiting environment and public health.

The work by De Vries (2021) calls for improvements in nitrogen use efficiency, particularly in agriculture, to reduce nitrogen run-off and emissions. He also advocates for interdisciplinary approaches that combine agricultural practices, environmental regulations, and public health initiatives to manage nitrogen impacts more sustainably.

The article by Nieder and Benbi (2021) demonstrates specific health risks associated with different nitrogen compounds. Nitrogen oxides (NO_x) are linked to respiratory issues and cardiovascular diseases when present in high levels in the air. Nitrate contamination (NO_3^-) in drinking water has been associated with methemoglobinemia and probable connections to positive cancers, especially in areas with high agricultural activity.

The paper by Panagos et al. (2022) discusses studies showing that long-term exposure to reactive nitrogen can increase risks of chronic diseases. The article highlights a need for a balanced phosphorus budget in European agricultural soils to prevent environmental degradation and maintain agricultural productivity. By adopting region-specific phosphorus management practices, Europe could enhance soil health, reduce water pollution, and promote sustainable agriculture.

The results by Makarenko et al. (2024) reveal that Finnish companies generally score higher on the Sustainability Transparency Index due to stricter regulations and a greater emphasis on sustainability reporting. In contrast, Ukrainian companies face challenges in implementing transparency measures, primarily due to less rigorous governance frameworks and fewer resources dedicated to sustainability initiatives.

The study by Dobrovolska et al. (2024) identifies several clusters of countries with similar agricultural practices, highlighting both developed and developing countries. For example, countries with high-tech, sustainable agricultural practices are grouped separately from countries with more traditional, resource-intensive farming models. These clusters reveal regional trends and provide insights into how local policies and market conditions shape agricultural business practices.

The research by Baidala et al. (2024) reflects substantial variation in development of organic farming across European countries. For instance, some countries like Germany, Austria, and Sweden have high percentages of agricultural land devoted to organic farming, driven by strong policy support and consumer demand. In contrast, other countries are still in early stages of organic farming adoption, primarily due to economic constraints or limited market demand.

The article by Zegunis and Balezentis (2024) finds that production risks in crop farming vary significantly between countries. For instance, the Baltic States experience higher yield variability due to less favorable climatic conditions, making their agricultural sector more susceptible to climate-induced risks. Germany and Poland, in contrast, benefit from more stable climatic conditions and better developed agricultural infrastructure, leading to lower production risks overall.

The study by Arych et al. (2023) runs a broad analysis of costs faced by agricultural holdings across the EU, including insurance as a key component of farming expenses. The findings underscore a need for policy measures that enhance access to affordable insurance and support sustainable farming practices to manage production risks effectively.

The article by Maity et al. (2023) provides valuable insights into the resource use efficiency of paddy production in West Bengal, highlighting a potential for cleaner agricultural practices. By adopting more efficient resource management techniques, paddy farms can improve productivity, reduce environmental impact, and support sustainable agriculture.

The paper by El Azhari et al. (2024) reveals that many farmers do not adhere to recommended fertilizer doses. Factors contributing to low adoption rates include limited access to information on the best practices, economic constraints (such as fertilizer cost), and lack of access to agricultural extension services. The findings indicate smaller farms and farmers with lower educational levels are less likely to adopt recommended practices.

The article by Ivale et al. (2024) provides valuable insights into adoption of precision agriculture technologies in Brazil, highlighting advantages, adoption disparities, and challenges faced by farmers. The findings underscore a need for targeted support to make precision agriculture technologies accessible to all farm sizes, which could improve agricultural productivity and sustainability in Brazil.

While there is substantial research relating fertilizer use to environmental pollution and health risks, there is also limited empirical evidence directly quantifying how these risks translate into healthcare costs across European countries. Most studies focus on either environmental impacts or health costs separately, rather than connecting the two in measurable ways. Research that establishes a direct causal link between fertilizer-induced environmental issues (water and air pollution) and health expenditure would enhance understanding of this relationship.

The aims of this research is to explore relationship between fertilizer consumption and health expenditure across European countries. Specifically, it investigates how use of fertilizers – particularly nitrogen-based fertilizers – may impact public health costs due to environmental and health risks, such as

nitrate pollution in water and GGEs. This research seeks to understand if regions with higher fertilizer use bear greater health expenditures due to pollution-related health issues. Additionally, it examines whether policies targeting sustainable agricultural practices could reduce health costs, thereby supporting both public health and environmental sustainability objectives.

Primary hypothesis: Higher fertilizer consumption correlates with increased health expenditure, as excessive fertilizer use contributes to environmental pollution, leading to public health issues that increase healthcare costs.

Secondary hypothesis: Import and export shares of food products may also influence health expenditure. Specifically:

- Higher food imports might increase health expenditures if imported food has lower safety standards or environmental costs;
- Increased food exports could affect health expenditure by potentially reducing domestic food quality or safety.

These hypotheses aim to assess if pollution and trade dynamics associated with fertilizer use lead to increased healthcare spending, providing insights for policy on sustainable agriculture to alleviate long-term healthcare costs.

METHODOLOGY

Panel regression analysis is a statistical method used to analyze data collected across multiple entities (in this case – countries) over time. This analysis allows for capturing both cross-sectional and time-series variations in the data, which helps account for heterogeneity across entities. In panel data analysis, three primary models are used:

- The Pooling Model;
- The Fixed Effect Model;
- The Random Effect Model.

Each model has specific assumptions and is applied depending on characteristics of the data and the research objective.

The sample of countries is formed with orientation on geographic locations (European countries and countries associated with EU) and data availability. The sample includes the following countries: Albania, Austria, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Denmark, Estonia, Finland, France, Georgia, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Moldova, the Netherlands, Norway, Poland, Portugal, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey, Ukraine, the United Kingdom. These countries are either members of the European Union, on the path to joining this union, or are associate members of this organization. Accordingly, their regulatory documentation for agricultural safety and application of mineral fertilizers is harmonized with EU directives. The European Union regulates use of mineral fertilizers within sustainable agriculture through directives and strategies aimed at reducing environmental impacts and ensuring the sustainable development of the agricultural sector.

The timeframe for data sampling was regulated by the maximum periods of data availability for all countries. Thus, in line with this principle, the time period has been defined as 2006-2021.

As a result, the panel data comprises 19 entities (countries) and 16 time periods.

The general form of a panel data model can be written as Formula (1) (Baltagi, 2005; Wooldridge, 2010; Hsiao, 2014; Greene, 2018):

$$y_{it} = \alpha_i + X_{it}\beta + u_{it} , \quad (1)$$

where y_{it} – a dependent variable for entity i at time t ; X_{it} – a matrix of independent variables for entity i at time t ; α_i – an entity-specific effect (individual heterogeneity); u_{it} – an idiosyncratic error term.

This research analyses the following variables:

- y_1 – current health expenditure per capita (PPP, current international \$) (World Bank, n.d.);
- y_2 – current health expenditure as a percentage of gross domestic product (GDP) (World Bank, n.d.);
- x_1 – fertilizer consumption (kilograms per hectare of arable land) (World Bank, n.d.);
- x_2 – import food product share (%) (WITS, n.d.);

- x_3 – export food product share (%) (WITS, n.d.).

Fertilizer consumption is the leading independent variable. The share of imported food products and the share of exported food products are added as control variables. The first control variable defines how many products were exported and, accordingly, did not enter the diet of the citizens of the respective country. Conversely, the second control variable determines how many products were imported and entered the diet of the citizens of the respective country. This means elements of the applied mineral fertilizers that could have entered the food products were exported. Vice versa, changes in the health of citizens may be associated with importing food products with mineral fertilizer residues from other countries.

The Pooling Model

The Pooling Model, also known as Pooled OLS, assumes there is no unobserved heterogeneity among entities (i.e., no individual effects). In other words, it suggests α_i is the same across all entities. The model can be specified as Formula (2) (Wooldridge, 2010; Hsiao, 2014; Greene, 2018):

$$y_{it} = \alpha_i + X_{it}\beta + u_{it} , \quad (2)$$

where y_{it} – a dependent variable for entity i at time t ; X_{it} – a matrix of independent variables for entity i at time t ; α_i – an entity-specific effect (individual heterogeneity); u_{it} – an idiosyncratic error term.

Estimation for this model is the Ordinary Least Squares (OLS). This model is based on following assumptions: homogeneity across entities α does not vary with i . The pooling model ignores the panel structure, treating the data as if it were a simple cross-sectional or time-series dataset. Limitation of this model is that the model may produce biased results if significant individual-specific effects exist.

The Fixed Effect Model

The Fixed Effect Model accounts for unobserved heterogeneity by assuming individual-specific effects α_i , which may correlate with the explanatory variables X_{it} . This model captures the time-invariant characteristics of each entity.

A common way to estimate this model is by removing α_i and via mean-differencing (within transformation). It results in Formula (3) (Baltagi, 2005; Wooldridge, 2010; Hsiao, 2014; Greene, 2018):

$$y_{it} - \bar{y}_i = (X_{it} - \bar{X}_i)\beta + (u_{it} - \bar{u}_i) , \quad (3)$$

where $\bar{y}_i$ and $\bar{X}_i$ are averages of y_{it} , and X_{it} is an average for each entity i over time.

Estimation is provided with the Within Estimator, which is an OLS type applied to the transformed equation. This model is based on the following assumptions: α_i is a fixed parameter potentially correlated with X_{it} . The Fixed Effect Model controls for all time-invariant factors, isolating the within-entity variation. However, this model does not estimate effects of time-invariant variables since they are removed in the transformation.

The Random Effect Model

The Random Effect Model assumes the unobserved individual effects α_i are random variables uncorrelated with the explanatory variables X_{it} . This assumption allows α_i to vary across entities, providing a more efficient estimate compared to the Fixed Effect Model if the assumption holds.

The Random Effect model can be specified as Formula (4) (Baltagi, 2005; Wooldridge, 2010; Hsiao, 2014; Greene, 2018):

$$y_{it} = \alpha + X_{it}\beta + \epsilon_{it} , \quad (4)$$

where $\epsilon_{it} = \alpha_i + u_{it}$.

Estimation is provided with the Generalized Least Squares (GLS), or Feasible GLS if the variance components are estimated. This model is based on following assumptions: α_i is uncorrelated with X_{it} and u_{it} is independently and identically distributed. The Random Effect Model implication allows for both within-entity and between-entity variations, improving efficiency if the Random Effect Model assumption is correct. There is a limitation of this model: if α_i is correlated with X_{it} , random effect estimates will be biased and inconsistent.

Model Selection

To decide between fixed effects and random effects, the Hausman test (1978) can be typically used:

- The null hypothesis: random effects are consistent and preferred;
- The alternative hypothesis: fixed effects are consistent, and random effects are inconsistent.

If the test is significant, the null hypothesis is rejected and the Fixed Effect Model is preferred. Otherwise, the Random Effect Model may be chosen.

RESULTS

Checking trends and heterogeneity within entities over time allows to get output of descriptive statistics, which is presented in Table 1. The data reveals substantial variation among countries in terms of health expenditure, fertilizer use, and share of food imports and exports. It reflects different economic and agricultural structures across the sample countries.

Table 1. Summarized Statistics of the Countries' Sample

Id	37					
Time	16					
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
y1	368.4	1370.9	2679.1	3045.8	4470.8	8998.4
y2	4.120	6.782	8.454	8.338	9.730	13.680
x1	0.00	94.69	133.18	173.98	191.25	1637.09
x2	0.4343	2.9061	4.8831	5.3273	6.8187	25.2604
x3	1.097	3.844	4.885	5.296	6.334	12.019

Source: author's calculations in R-Studio (2024)

Note: Min. (Minimum) – the smallest value in the dataset; 1st Qu. (First Quartile, or 25th percentile) – the value below which 25% of the data fall; Median (50th percentile) – the middle value of the dataset, where half of the data are below and half are above; Mean – the average of all values in the dataset; 3rd Qu. (Third Quartile, or 75th percentile) – three-fourths of the data; Max. (Maximum) – the largest value in the dataset.

There is a wide range of health expenditures per capita, with variation among countries, as indicated by a large difference between the minimum (\$368.4) and the maximum (\$8998.4). The mean (3045.8) is higher than the median (2679.1), suggesting a right-skewed distribution, where a few countries have very high health expenditures per capita. Health expenditure as a percentage of GDP is more tightly clustered compared to per capita spending, with most values being between 4.12% and 13.68%. The mean and median are close, indicating a roughly symmetric distribution. Fertilizer consumption varies widely among countries, with a few countries having very high usage levels. The mean is higher than the median, indicating that the distribution is right-skewed, likely due to a few countries with extremely high fertilizer consumption. The import food product share also has a wide range, with a few countries having very high shares of imported food products, as indicated by the maximum value (25.26%). The mean is slightly higher than the median, suggesting a slight right skew. The export food product share is more concentrated, with most values between 1.097% and 12.019%. The mean and the median are close, indicating a roughly symmetric distribution.

Table 2 provides an output from the Pooling Model analysis in R, which examines relationships between current health expenditure per capita (y1) as the dependent variable and three independent variables: fertilizer consumption (x1), import food product share (x2), and export food product share (x3). The data set is balanced, meaning each entity (in this case, likely countries) has the same number of time observations. There are 37 entities, each observed over 16 time periods, resulting in a total of 592 observations.

Residuals indicate spread of errors in predicting y1 and suggest some variability in the residuals, with a few large errors (positive and negative). It indicates that the model may not capture all factors affecting y1.

Each coefficient provides insight into direction and strength of relationship between the independent variables and y1. The baseline value of current health expenditure per capita has an estimation coefficient for intercept of 3435.6226 when all independent variables are zero. While it is unlikely that all variables are actually zero in practice, this provides a baseline level for comparison. The coefficient of x1 (fertilizer consumption) is highly significant (p-value < 2.2e-16) with a positive relationship (estimate = 3.6134) between fertilizer consumption and current health expenditure per capita. For each additional kilogram of fertilizer per hectare of arable land, y1 is expected to increase by approximately 3.61 units, holding other factors constant.

Table 2. Pooling Model Output

Pooling Model				
Call:				
plm(formula = y1 ~ x1 + x2 + x3, data = pdata, model = "pooling")				
Balanced Panel: n = 37, T = 16, N = 592				
Residuals:				
Min.	1st Qu.	Median	3rd Qu.	Max.
-2840.38	-1348.57	-263.48	1140.30	5403.00
Coefficients:				
Variables	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	3435.6226	199.9231	17.1847	< 2.2e-16***
x1	3.6134	0.3676	9.8297	< 2.2e-16***
x2	-97.6836	22.4725	-4.3468	1.628e-05***
x3	-94.0393	39.4767	-2.3821	0.01753*
Signif. codes: '***' – 0.001; '**' – 0.01; '*' – 0.05; '.' – 0.1 ' ' – 'No symbol' – insignificant.				
Total Sum of Squares: 2123300000				
Residual Sum of Squares: 1720300000				
R-Squared: 0.18981				
Adj. R-Squared: 0.18567				
F-statistic: 45.9173 on 3 and 588 DF, p-value: < 2.22e-16				

Source: author's calculations in R Studio (2024)

Note: Min. (Minimum) – the smallest value in the dataset; 1st Qu. (First Quartile, or 25th percentile) – the value below which 25% of the data fall; Median (50th percentile) – the middle value of the dataset, where half of the data are below and half are above; Mean – the average of all values in the dataset; 3rd Qu. (Third Quartile, or 75th percentile) – three-fourths of the data; Max. (Maximum) – the largest value in the dataset; Estimate – the estimated coefficient for a predictor variable; Std. Error (Standard Error) – measures the precision of the estimated coefficient; t-value – the statistic is calculated as the ratio of the estimate to the standard error (t-value = Estimate / Std. Error); Pr(>|t|) (p-value) – the probability of observing a t-value as extreme as, or more extreme than, the one calculated, assuming the null hypothesis is true.

The variable x2 (import food product share) is also significant (p-value = 1.628e–05). It has a negative coefficient (-97.6836) suggesting that a higher share of food product imports is associated with a decrease in current health expenditure per capita. For each percentage increase in food imports, y1 decreases by approximately 97.68 units, assuming other factors are constant.

The variable x3 (export food product share) is significant at the 0.05 level (p-value = 0.01753). Its negative coefficient (-94.0393) suggests an increase in food export share is also associated with a decrease in current health expenditure per capita. For each percentage point increase in food exports, y1 decreases by approximately 94.04 units, holding other factors constant.

R-squared = 0.18981 means that about 18.98% of the variability in y1 is explained by the model. This relatively low R-squared suggests that other factors outside this model could also influence current health expenditure per capita. This adjusted value (0.18567) accounts for a number of predictors and is slightly lower than R-squared. It is common when adding more variables does not substantially increase the explanatory model power.

The F-statistic (45.9173) tests the overall model significance. The very low p-value (<2.22e–16) indicates that at least one of the predictors (x1, x2, x3) has a statistically significant relationship with y1.

This model suggests that while fertilizer consumption is associated with higher health expenditure per capita, both import and export shares of food products are associated with lower health expenditure per capita. This could point to complex relationships between agriculture, trade, and health expenditure.

Output from the one-way (individual) effect within model for panel data R analysis is provided in Table 3. This is a balanced panel data set with 37 entities (possibly countries or regions), each observed over 16 periods, resulting in a total of 592 observations.

Table 3. One-Way Fixed Effects (within) Model Output

Oneway (individual) effect Within Model				
Call: plm(formula = y1 ~ x1 + x2 + x3, data = pdata, model = "within")				
Unbalanced Panel: n = 37, T = 16, N = 592				
Residuals:				
Min.	1st Qu.	Median	3rd Qu.	Max.
-2131.225	-322.516	-57.332	279.094	24497.966
Coefficients:				
Variable	Estimate	Std. Error	t-value	Pr(> t)
x1	2.61814	0.56281	4.6519	4.122e-06 ***
x2	11.51524	21.00288	0.5483	0.5837
x3	381.80062	32.69133	11.6790	< 2.2e-16 ***
Total Sum of Squares: 296130000				
Residual Sum of Squares: 221990000				
R-Squared: 0.25036				
Adj. R-Squared: 0.19739				
F-statistic: 61.4499 on 3 and 552 DF, p-value: < 2.22e-16				

Source: author's calculations in R Studio (2024)

Note: Min. (Minimum) – the smallest value in the dataset; 1st Qu. (First Quartile, or 25th percentile) – the value below which 25% of the data fall; Median (50th percentile) – the middle value of the dataset, where half of the data are below and half are above; Mean – the average of all values in the dataset; 3rd Qu. (Third Quartile, or 75th percentile) – three-fourths of the data; Max. (Maximum) – the largest value in the dataset; Estimate – the estimated coefficient for a predictor variable; Std. Error (Standard Error) – measures the precision of the estimated coefficient; t-value – the statistic is calculated as the ratio of the estimate to the standard error (t-value = Estimate / Std. Error); Pr(>|t|) (p-value) – the probability of observing a t-value as extreme as, or more extreme than, the one calculated, assuming the null hypothesis is true.

The residuals provide information on the spread of prediction errors for y1. They suggest a range of residuals from large negative to large positive, indicating there is some variability in how well the model captures y1 values for each observation.

x1 (fertilizer consumption) is highly significant (p-value = 4.122e-06), indicating a positive relationship (2.61814) between fertilizer consumption and current health expenditure per capita. For each additional kilogram of fertilizer per hectare, y1 is expected to increase by approximately 2.62 units, holding other factors constant.

x2 (import food product share) is not statistically significant (p-value = 0.5837), indicating that import share of food products does not have a meaningful effect on y1 in this model.

x3 (export food product share) is statistically significant at the highest level (p-value < 2.2e-16), showing a strong positive relationship with y1. For each percentage increase in food exports, y1 increases by about 381.80 units, holding other factors constant.

Approximately 25.04% of the y1 variation is explained by the model (R-squared = 0.25036). While the model captures some of the variability, it suggests there are other factors influencing y1 not included in the model. This adjusted R-squared (0.19739) accounts for a range of predictors and indicates a slightly lower fit. The F-statistic (61.4499) tests the overall model significance. The very low p-value (<2.22e-16) indicates the model, as a whole, is statistically significant, meaning at least one of predictors (x1, x2, x3) is related to y1.

This result suggests fertilizer consumption and export share of food products are positively associated with health expenditure per capita. Import share of food products does not have a significant effect in this model.

Table 4 provides output from the Random Effect Model (via the Swamy-Arora transformation) in R. This analysis examines relationships between current health expenditure per capita, PPP (y1) as the dependent variable, and three independent variables: fertilizer consumption (x1), import food product share (x2), and export food product share (x3). This is also a balanced panel with 37 entities, each observed over 16 time periods, for a total of 592 observations.

Idiosyncratic variance (402155.8) with a standard deviation of 634.26 and a share of 0.139 represents variance due to random noise (idiosyncratic error) in the model. Individual variance (2488852.5) with a standard deviation of 1577.6 and a share of 0.861 represents variance attributed to individual differences among the entities. Theta (0.9) is a weight parameter used in the random effect transformation, indicating relative influence of individual (entity-specific) effects in the model. The residuals indicate spread of errors

for y1 predictions and show some variability, with residual spread across a wide range.

Table 4. One-Way (Individual) Random Effect Model Output

Oneway (individual) effect Random Effect Model (Swamy-Arora transformation) Call: plm(formula = y1 ~ x1 + x2 + x3, data = pdata, model = "random")				
Balanced Panel: n = 37, T = 16, N = 592				
Effects:				
	var	std.dev	share	
idiosyncratic	402155.8	634.2	0.139	
individual	2488852.5	1577.6	0.861	
theta	0.9			
Residuals:				
Min.	1st Qu.	Median	3rd Qu.	Max.
-1699.89	-381.99	-99.45	259.26	2858.97
Coefficients:				
Variable	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	677.31917	325.90695	2.0783	0.03769 *
x1	2.65985	0.52999	5.0187	5.202e-07 ***
x2	1.55057	20.79689	0.0746	0.94057
x3	358.27706	32.57662	10.9980	< 2.2e-16 ***
Signif. codes: '***' – 0.001; '**' – 0.01; '*' – 0.05; '.' – 0.1 ' ' – 'No symbol' – insignificant.				
Total Sum of Squares: 314400000				
Residual Sum of Squares: 244950000				
R-Squared: 0.22088				
Adj. R-Squared: 0.2169				
Chisq: 166.694 on 3 DF, p-value: < 2.22e-16				

Source: author's calculations in R Studio (2024)

Note: Min. (Minimum) – the smallest value in the dataset; 1st Qu. (First Quartile, or 25th percentile) – the value below which 25% of the data fall; Median (50th percentile) – the middle value of the dataset, where half of the data are below and half are above; Mean – the average of all values in the dataset; 3rd Qu. (Third Quartile, or 75th percentile) – three-fourths of the data; Max. (Maximum) – the largest value in the dataset; Estimate – the estimated coefficient for a predictor variable; Std. Error (Standard Error) – measures the precision of the estimated coefficient; t-value – the statistic is calculated as the ratio of the estimate to the standard error (t-value = Estimate / Std. Error); Pr(>|t|) (p-value) – the probability of observing a t-value as extreme as, or more extreme than, the one calculated, assuming the null hypothesis is true.

Interpretation (677.31917) is the expected baseline value of y1 (current health expenditure per capita) when all independent variables are zero. Though it is unlikely that all variables are actually zero, this value serves as a baseline. x1 (fertilizer consumption) with the estimate (2.65985) is highly significant (p-value = 5.202e-07), indicating a positive relationship between fertilizer consumption and current health expenditure per capita. For each additional kilogram of fertilizer per hectare, y1 is expected to increase by approximately 2.66 units, holding other factors constant.

x2 (import food product share) is not statistically significant (p-value > 0.05), indicating the import share of food products does not have a meaningful effect on y1 in this Random Effect Model.

x3 (export food product share) is highly significant (p-value < 2.2e-16), showing a strong positive relationship with y1. For each percentage increase in the export share of food products, y1 is expected to increase by about 358.28 units, holding other factors constant.

R-squared (0.22088) means approximately 22.09% of the y1 variation is explained by the model, indicating moderate explanatory power. The adjusted value (0.2169) accounts for a range of predictors, showing a similar level of fit. The chi-square test (166.694 with 3 degrees of freedom) indicates the overall model is statistically significant, suggesting at least one of the predictors (x1, x2, x3) has a significant effect on y1.

This model demonstrates factors related to fertilizer consumption and food export shares may positively impact health expenditure per capita while food import shares do not appear to have a significant effect in this analysis.

The research uses the Hausman test (1978) to define whether the Fixed Effect Model or the Random

Effect Model is more appropriate for panel data analysis. Since the p-value is extremely small ($< 2.2e-16$, significantly below the typical 0.05 threshold), the null hypothesis is rejected in favor of the alternative hypothesis. The Hausman test suggests the Fixed Effect Model is more appropriate than the Random Effect Model for input data. This indicates there is a correlation between the individual (entity-specific) effects and the independent variables x_1 , x_2 , x_3 . Using the Random Effect Model would likely lead to biased estimates.

Table 5 provides output from the Pooling Model analysis in R. It examines relationships between current health expenditure as a GDP percentage (y_2 , a dependent variable) and three independent variables: fertilizer consumption (x_1), import food product share (x_2), and export food product share (x_3). This is a balanced panel data set with 37 entities (countries). Each of them is observed over 16 periods, resulting in 592 observations.

Table 5. Pooling Model Output

Pooling Model				
Call:				
plm(formula = $y_2 \sim x_1 + x_2 + x_3$, data = pdata, model = "pooling")				
Balanced Panel: n = 37, T = 16, N = 592				
Residuals:				
Min.	1st Qu.	Median	3rd Qu.	Max.
-4.05494	-1.59220	0.14229	1.43639	4.75146
Coefficients:				
Variable	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	7.92099370	0.21640512	36.6026	$< 2e-16$ ***
x_1	0.00085925	0.00039791	2.1594	0.03122 *
x_2	0.04809257	0.02432523	1.9771	0.04850 *
x_3	0.00220495	0.04273123	0.0516	0.95886
Signif. codes: '***' – 0.001; '**' – 0.01; '*' – 0.05; '.' – 0.1; ' ' – 'No symbol' – insignificant.				
Total Sum of Squares: 2049.5				
Residual Sum of Squares: 2015.7				
R-Squared: 0.016508				
Adj. R-Squared: 0.01149				
F-statistic: 3.28983 on 3 and 588 DF, p-value: 0.020383				

Source: author's calculations in R Studio (2024)

Note: Min. (Minimum) – the smallest value in the dataset; 1st Qu. (First Quartile, or 25th percentile) – the value below which 25% of the data fall; Median (50th percentile) – the middle value of the dataset, where half of the data are below and half are above; Mean – the average of all values in the dataset; 3rd Qu. (Third Quartile, or 75th percentile) – three-fourths of the data; Max. (Maximum) – the largest value in the dataset; Estimate – the estimated coefficient for a predictor variable; Std. Error (Standard Error) – measures the precision of the estimated coefficient; t-value – the statistic is calculated as the ratio of the estimate to the standard error ($t\text{-value} = \text{Estimate} / \text{Std. Error}$); Pr(>|t|) (p-value) – the probability of observing a t-value as extreme as, or more extreme than, the one calculated, assuming the null hypothesis is true.

Most residuals are clustered near zero, but with a few larger positive and negative residuals.

The baseline y_2 value is 7.92099370 with all independent variables being zero. While it is unlikely for all variables to be zero, this serves as a starting point for interpreting other coefficients.

x_1 (fertilizer consumption) is significant at the 0.05 level ($p\text{-value} = 0.03122$), indicating a small positive relationship (0.00085925) between fertilizer consumption and health expenditure as a percentage of GDP. For each additional kilogram of fertilizer per hectare, y_2 is expected to increase by 0.00086 percentage points, holding other factors constant. x_2 is also significant at the 0.05 level ($p\text{-value} = 0.04850$), suggesting a positive (0.04809257) relationship between share of food imports and health expenditure as a percentage of GDP. For each percentage point increase in import share of food products, y_2 is expected to rise by approximately 0.048 percentage points, assuming other factors are constant.

x_3 (export food product share) is not statistically significant ($p\text{-value} = 0.95886 > 0.05$), indicating export share of food products does not have a meaningful effect on y_2 in this model.

R-squared (0.016508) means approximately 1.65% of the y_2 variation is explained by the model. It suggests other factors not included in this model likely play a significant role in influencing current health expenditure as a percentage of GDP. The adjusted R-squared (0.01149) is even lower, further indicating this model has a limited explanatory power. The F-statistic (3.28983) tests the overall model significance. The p-value of 0.020383 is below the 0.05 threshold, meaning the model is statistically significant as a whole. However, the low R-squared suggests while the model is statistically significant, it does not capture much of

the y_2 variance.

This model suggests while fertilizer consumption and food import share have a statistically significant, albeit small, positive effect on health expenditure as a percentage of GDP, export share of food products does not significantly impact health expenditure in this case. The overall model, while statistically significant, has a limited explanatory power.

Output from the one-way (individual) effect within model in R (Table 6) examines relationships between y_2 as the dependent variable and fertilizer consumption (x_1), import food product share (x_2), and export food product share (x_3) as independent variables. The residuals show most predictions are close to actual values, with some larger deviations in both directions.

Table 6. One-Way Fixed Effects (within) Model Output

Oneway (individual) effect Within Model				
Call: <code>plm(formula = y2 ~ x1 + x2 + x3, data = pdata, model = "within")</code>				
Unbalanced Panel: n = 37, T = 16, N = 592				
Residuals:				
Min.	1st Qu.	Median	3rd Qu.	Max.
-1.935906	-0.359151	-0.055377	0.309032	3.983080
Coefficients:				
Variable	Estimate	Std. Error	t-value	Pr(> t)
x_1	-0.00336508	0.00058624	-5.7401	1.562e-08 ***
x_2	0.07362926	0.02187715	3.3656	0.0008169 ***
x_3	0.35962171	0.03405214	10.5609	< 2.2e-16 ***
Total Sum of Squares: 332.84				
Residual Sum of Squares: 240.86				
R-Squared: 0.27636				
Adj. R-Squared: 0.22523				
F-statistic: 70.2687 on 3 and 552 DF, p-value: < 2.22e-16				

Source: author's calculations in R Studio (2024)

Note: Min. (Minimum) – the smallest value in the dataset; 1st Qu. (First Quartile, or 25th percentile) – the value below which 25% of the data fall; Median (50th percentile) – the middle value of the dataset, where half of the data are below and half are above; Mean – the average of all values in the dataset; 3rd Qu. (Third Quartile, or 75th percentile) – three-fourths of the data; Max. (Maximum) – the largest value in the dataset; Estimate – the estimated coefficient for a predictor variable; Std. Error (Standard Error) – measures the precision of the estimated coefficient; t-value – the statistic is calculated as the ratio of the estimate to the standard error ($t\text{-value} = \text{Estimate} / \text{Std. Error}$); $\text{Pr}(>|t|)$ (p-value) – the probability of observing a t-value as extreme as, or more extreme than, the one calculated, assuming the null hypothesis is true.

All variables are highly significant:

x_1 (fertilizer consumption) is -0.00336508 with $p\text{-value} = 1.562\text{e-}08$, indicating a negative relationship between fertilizer consumption and health expenditure. For each additional kilogram of fertilizer per hectare, y_2 is expected to decrease by 0.0034 percentage points, holding other factors constant within $0.00336508\text{--}0.00336508$.

x_2 (import food product share) with $p\text{-value} = 0.0008169$ has a positive relationship between share of food imports and health expenditure. For each percentage increase in import share of food products, y_2 is expected to increase by approximately 0.0736 percentage points, holding other factors constant.

x_3 (export food product share) with $p\text{-value} < 2.2\text{e-}16$ positively contributes to y_2 . For each percentage increase in export share of food products, y_2 is expected to increase by approximately 0.3596 percentage points, holding other factors constant.

Approximately 27.64% of the y_2 variation is explained by the model. This indicates a moderate level of explanatory power, suggesting other factors outside this model may also influence health expenditure. The adjusted R-squared accounts for a range of predictors and provides a slightly lower fit value. Still, it shows moderate explanatory power. The F-statistic tests the overall model significance. The very low p-value indicates the model as a whole is statistically significant, meaning at least one of the predictors (x_1 , x_2 , x_3) has a significant relationship with y_2 .

This model suggests while fertilizer consumption is associated with a small decrease in health expenditure as a percentage of GDP, both import and export shares of food products are associated with an increase in this expenditure. The export share shows a substantial positive association, potentially indicating an economic relationship between trade in food products and health spending patterns.

Output from the one-way (individual) Random Effect Model (via the Swamy-Arora transformation) in R for y2 is presented in Table 7. These residuals indicate some variability around predictions, with most errors clustered near zero, but a few larger deviations in both directions. Theta close to 1 suggests that individual effects have a strong impact in this model.

Table 7. One-Way (Individual) Random Effect Model Output

Oneway (individual) effect Random Effect Model (Swamy-Arora transformation) Call: plm(formula = y1 ~ x1 + x2 + x3, data = pdata, model = "random")				
Balanced Panel: n = 37, T = 16, N = 592				
Effects:				
	var	std.dev	share	
idiosyncratic	0.4363	0.6606	0.122	
individual	3.1372	1.7712	0.878	
theta	0.9072			
Residuals:				
Min.	1st Qu.	Median	3rd Qu.	Max.
-2.171026	-0.380607	-0.044851	0.323345	3.895799
Coefficients:				
Variable	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	6.62908802	0.35461861	18.694	< 2.2e-16***
x1	-0.00287942	0.00055288	-5.208	1.909e-07***
x2	0.06866279	0.02155851	3.185	0.001448**
x3	0.34825966	0.03374089	10.322	< 2.2e-16***
Signif. codes: '***' – 0.001; '**' – 0.01; '*' – 0.05; '.' – 0.1 ' ' – 'No symbol' – insignificant.				
Total Sum of Squares: 347.63				
Residual Sum of Squares: 261.29				
R-Squared: 0.24838				
Adj. R-Squared: 0.24454				
Chisq: 194.306 on 3 DF, p-value: < 2.22e-16				

Source: author's calculations in R Studio (2024)

Note: Min. (Minimum) – the smallest value in the dataset; 1st Qu. (First Quartile, or 25th percentile) – the value below which 25% of the data fall; Median (50th percentile) – the middle value of the dataset, where half of the data are below and half are above; Mean – the average of all values in the dataset; 3rd Qu. (Third Quartile, or 75th percentile) – three-fourths of the data; Max. (Maximum) – the largest value in the dataset; Estimate – the estimated coefficient for a predictor variable; Std. Error (Standard Error) – measures the precision of the estimated coefficient; t-value – the statistic is calculated as the ratio of the estimate to the standard error (t-value = Estimate / Std. Error); Pr(>|t|) (p-value) – the probability of observing a t-value as extreme as, or more extreme than, the one calculated, assuming the null hypothesis is true.

The baseline y2 level is expected when all independent variables are zero (Intercept = 6.62908802). While it is unlikely for all variables to be zero, this value serves as a baseline reference.

x1 (fertilizer consumption) is highly significant (p-value = 1.909e-07), indicating a small but statistically significant negative (-0.00287942) relationship between fertilizer consumption and health expenditure. For each additional kilogram of fertilizer per hectare, y2 is expected to decrease by approximately 0.0029 percentage points, holding other factors constant. x2 is also highly significant (p-value = 0.00144). It has a positive relationship between share of food imports and health expenditure. For each percentage increase in import share of food products, y2 is expected to increase by approximately 0.0687 percentage points, holding other factors constant. x3 is also highly significant (p-value < 2.2e-16), showing a strong positive relationship between export share of food products and health expenditure. For each percentage increase in the export share, y2 is expected to rise by about 0.3483 percentage points, holding other factors constant.

Approximately 24.84% of the y2 variation is explained by this model. While the model captures some of the health expenditure variability, there may still be other factors that are not included here. This adjusted R-squared value, which accounts for a range of predictors, is slightly lower, suggesting a similar fit. The chi-square test indicates the overall model is statistically significant, meaning at least one of predictors (x1, x2, x3) has a significant effect on y2.

This result suggests while fertilizer consumption is associated with a small decrease in health

expenditure, higher import and export shares of food products are associated with increased health expenditure, with export share showing a relatively large positive effect. This could imply a complex relationship between agricultural trade, domestic food security, and health spending patterns in the observed entities.

The Hausman test is used to decide whether the Fixed Effect Model or the Random Effect Model is more appropriate for panel data analysis. The Hausman test results are following:

- Chi-square statistic (chisq) = 23.866;
- Degrees of freedom (df) = 3 (there are 3 independent variables);
- p-value = 2.665×10^{-5} (0.00002665);
- Alternative hypothesis: one model is inconsistent.

The Hausman test indicates the Fixed Effect Model is more appropriate for making inferences about relationships between y2 (current health expenditure) and fertilizer consumption, import food product share, and export food product share. The Fixed Effect Model controls for unobserved heterogeneity that could influence y2, resulting in more reliable and consistent estimates.

DISCUSSION

A positive effect of fertilizer consumption on health expenditure suggests that:

- Higher fertilizer usage may be associated with more intensive agricultural practices, possibly requiring more workforce, resources, or infrastructure in the agricultural sector. This can indirectly influence health expenditure because labor-intensive agricultural practices can lead to increased health risks and occupational hazards, demanding greater health support;
- Excessive or higher fertilizer usage can sometimes have adverse environmental effects, such as water contamination or soil degradation. These environmental factors can indirectly affect public health, potentially increasing healthcare costs to manage pollution-related health issues;
- Countries with higher fertilizer consumption might be in a phase of economic development where agricultural productivity is a priority. Higher productivity can increase income, some of which might be allocated to health spending. Additionally, more productive agricultural economies may support better public health infrastructure.

A positive effect of food import share on health expenditure might imply:

- Countries that rely more on imported food may face higher food costs and less control over food quality and safety. Increased food import share could be linked to health expenditures if imported foods are less regulated or if there are health impacts related to processed or low-quality food imports;
- Countries with a higher share of food imports might also be more economically developed, as many economically advanced nations import a large portion of their food. In such cases, higher economic stability could mean that governments can afford to spend more on healthcare, increasing health expenditure;
- Countries reliant on food imports may be more susceptible to international market fluctuations or supply chain disruptions, which can impact food security. A focus on food security could lead to increased health spending to ensure access to adequate nutrition and healthcare during disruptions.

The result that fertilizer consumption is associated with a small decrease in health expenditure as a percentage of GDP might seem counterintuitive at first, but a possible explanation could be the following. When a country's agricultural sector becomes more efficient (through higher yields due to increased fertilizer use), it may result in a country's GDP rise because export of agricultural products grows. However, this could also mean the relative percentage of GDP spent specifically on healthcare falls as a larger portion of GDP is produced by the agricultural sector.

The discrepancy between Fixed and Random Effect Models suggests there is an omitted variable or unobserved characteristic that affects relationships between the independent variable and the dependent variable in a systematic way. This unobserved factor is likely correlated with the independent variable in question, which causes the Random Effect Model to capture a different effect than the Fixed Effect Model. In the Random Effect Model, if an unobserved factor (GDP) affects both the dependent and independent variables, it can bias the coefficient. Since the Random Effect Model does not control for these unobserved entity-specific factors, it may overestimate or underestimate the relationships, leading to a positive result.

The Fixed Effect Model only looks at how changes within each entity over time are associated with changes in the dependent variable. In contrast, the Random Effect Model also incorporates variation between entities. The positive outcome in the Random Effect Model might be due to between-entity differences that the Fixed Effect Model eliminates. For example, if countries with higher values of the independent variable generally have better health outcomes due to other characteristics (higher income, better infrastructure), the Random Effect Model might capture this positive effect. However, within each country, increases in this variable might have a negative impact, which is what the Fixed Effect Model would detect.

If the Fixed Effect Model is more appropriate (based on the Hausman test), it suggests policies aiming to raise the independent variable within a given country may have a different effect than policies that compare levels of this variable across countries. The negative coefficient in the Fixed Effect Model indicates any conclusions drawn from the Random Effect Model should be treated cautiously because they may be influenced by omitted variables correlated with the independent variable.

This research provides timely and critical insights into relationships between fertilizer consumption and health expenditure in European countries, highlighting the intersection of agricultural practices, environmental sustainability, and public health. As environmental concerns grow due to adverse impacts of nitrogen-based fertilizers (such as water contamination, GGEs, and soil degradation), understanding these effects on public health and healthcare spending becomes increasingly urgent. The study addresses these pressing concerns by investigating whether higher fertilizer consumption in Europe is associated with increased health expenditure, considering how pollution and trade in food products might shape health costs in various countries.

The methodology employed in this research is rigorous and well-suited to the complexity of the topic. Via panel regression analysis across 37 European countries from 2006 to 2021, the study applies a range of statistical models (the Pooling, Fixed Effect, and Random Effect Models) to control for country-specific and time-invariant variables. This approach allows for a nuanced understanding of how fertilizer consumption, and import and export shares of food products relate to health expenditure per capita and as a percentage of GDP. The Hausman test was used to identify the most appropriate model for the data, ultimately favoring the Fixed Effect Model due to its ability to capture unobserved heterogeneity across countries. This methodological rigor enhances the study reliability, providing more accurate estimates of how agricultural practices influence health spending.

The research results reveal nuanced relationships between fertilizer consumption, food trade, and health expenditure across European countries. They provide critical insights into economic and health impacts of agricultural practices. Specifically:

- The analysis confirms a significant positive association between fertilizer consumption and health expenditure per capita. For each additional kilogram of fertilizer used per hectare, health spending rises. This finding supports the hypothesis that higher fertilizer usage – particularly nitrogen-based fertilizers – contributes to environmental pollution, such as nitrate contamination in water sources and GGEs. These environmental impacts are linked to public health issues, including respiratory problems, certain cancers, and conditions like methemoglobinemia (blue baby syndrome). Thus, countries with more intensive fertilizer use face higher healthcare costs associated with managing pollution-related health conditions;
- The results also indicate a positive relationship between food import share and health expenditure. This suggests that countries with higher reliance on imported food may incur increased health costs, possibly due to lower control over food quality and safety standards in imports. Imported food products may carry residues from fertilizers or pesticides used in the source countries, adding potential health risks. Additionally, greater food imports may reflect broader economic development and wealth, potentially leading to more comprehensive healthcare spending;
- The relationship between food exports and health expenditure reveals more complexity. In the Fixed Effect Model, higher export shares were associated with a health expenditure fall per capita, possibly suggesting economic benefits that support public health infrastructure. However, in other models, food product export also showed a positive association, indicating countries with high agricultural exports may face increased costs due to pollution or health effects associated with intensive agricultural practices. This complexity underscores the role of trade dynamics in shaping health outcomes. Simultaneously, export-focused agricultural economies might benefit economically yet still encounter higher health-related expenses from environmental costs of intensified farming;
- The study employs a range of statistical models to ensure the robustness of these results. The Fixed Effect Model was ultimately favored, as confirmed by the Hausman test, due to its ability to control for

unobserved heterogeneity within each country over time. This choice provides confidence that the observed relationships between fertilizer use, trade in food products, and health expenditure are not due to omitted variables or unobserved factors that vary between countries. The relatively low R-squared values in several models indicate that, while fertilizer consumption and trade dynamics play a significant role, other factors not included in the analysis may also influence health spending trends;

- The study results highlight the economic burden that environmental pollution from fertilizers places on public health systems. Regions with high fertilizer use not only face immediate environmental degradation but also bear longer-term health costs, supporting the need for sustainable agriculture that minimizes reliance on synthetic fertilizers. Moreover, the findings imply food trade policies also impact national health costs, as both import and export dynamics can influence health outcomes and expenditures.

The findings underline the importance of sustainable agricultural policies and the potential for preventative measures to mitigate healthcare costs associated with pollution. As European countries aim to balance agricultural productivity with environmental sustainability, this research offers valuable evidence to guide policy. The positive relationship between fertilizer application and health expenditure suggests targeted interventions (such as fertilizer run-off reduction and organic or precision farming encouragement) could decrease healthcare burdens over time. In a context where achieving sustainability goals is paramount, this study contributes a crucial perspective on how agricultural practices can support or hinder public health. It can also provide an essential consideration for European policy frameworks (including the European Green Deal and Sustainable Development Goals).

Author Contributions

Conceptualization: M. G.; data curation: M. G.; formal analysis: M. G.; investigation: M. G.; methodology: M. G.; project administration M. G.; supervision: M. G.; validation: M. G.; visualization: M. G.; writing – original draft: M. G.; writing – review & editing: M. G.

Conflict of Interest

The author declares no conflicts of interest.

Data Availability Statement

Not applicable.

Informed Consent Statement

Not applicable.

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